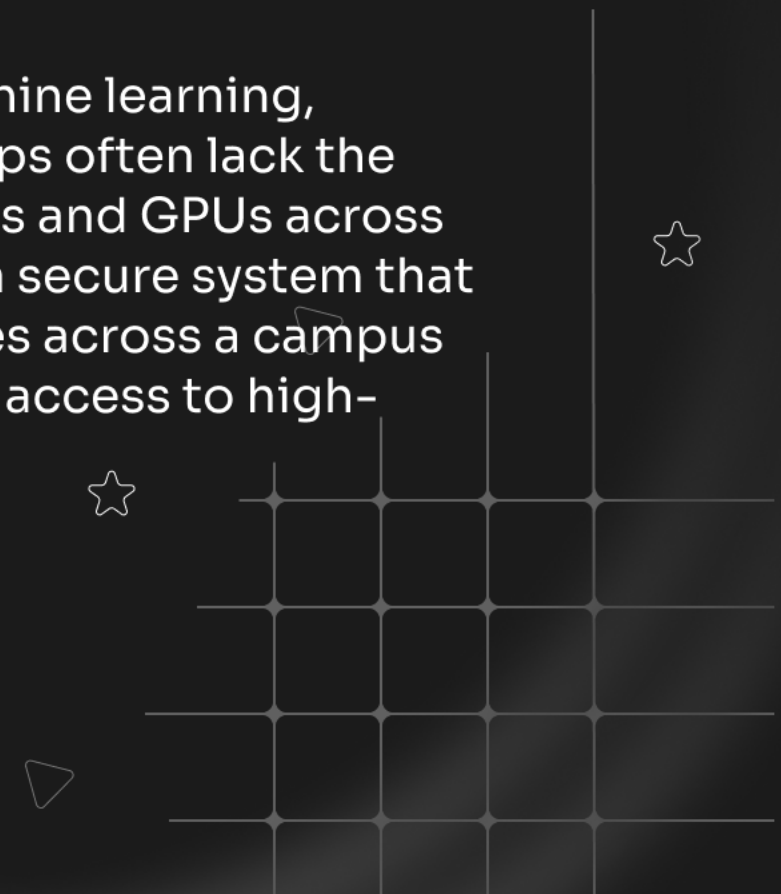


Problem Statements



Students often require high computing power for tasks like machine learning, simulations, and large-scale data processing, but personal laptops often lack the performance needed for these workloads. Meanwhile, many CPUs and GPUs across campus remain idle for long periods. The challenge is to design a secure system that allows students to share and access unused computing resources across a campus network, enabling efficient workload distribution and improving access to high-performance computing.



A Voice Agent allows users to interact with an AI through spoken conversation instead of text. The typical pipeline includes speech-to-text (STT) to convert audio into text, a language model that processes the request, and text-to-speech (TTS) that converts the response back into audio. In this challenge, you will build a real-time Voice Agent powered by one or more fine-tuned Small Language Models (under ~10B parameters). Because SLMs are smaller and faster, they can reduce latency and run efficiently on edge devices, making them ideal for responsive voice-based systems.

Participants must use Oumi to fine-tune the models and build a solution that addresses a realistic use case.

[Click here for detailed explanation](#)



ID- EC602

Sponsor Track

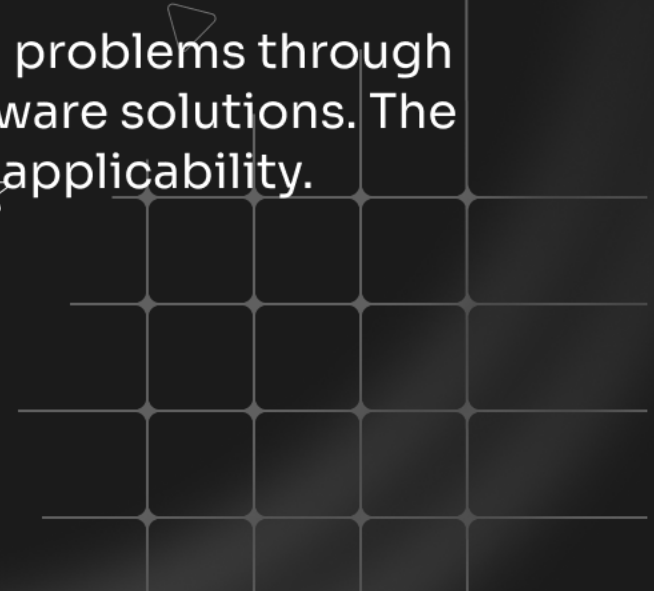
INR 10k PRIZE

Software

Focus on building innovative software solutions such as applications, platforms, AI systems, or digital tools that address real-world problems. Solutions may include web, mobile, cloud, or AI-based technologies that improve efficiency, accessibility, or user experience.

Hardware

Design and develop physical or embedded systems that solve practical problems through electronics, sensors, IoT devices, robotics, or integrated hardware–software solutions. The focus is on creating functional prototypes that demonstrate real-world applicability.



Modern organizations store large amounts of information across documents, databases, and internal tools, but this knowledge is often fragmented across disconnected systems. As a result, employees spend significant time searching through multiple platforms or relying on colleagues to find information that already exists. This slows productivity and leads to duplicated effort and missed insights. The challenge is to design solutions that enable efficient discovery and retrieval of relevant knowledge across enterprise systems while maintaining context and ensuring responses are grounded in reliable organizational data.



Survivors of sexual violence, torture, trafficking, and domestic abuse often face legal systems that require precise, linear accounts of traumatic events under intense scrutiny. However, trauma frequently causes fragmented memory, emotional distress, and inconsistent recall—responses that are clinically normal but often interpreted in court as signs of unreliability. Survivors must also repeat their experiences across multiple institutions, increasing re-traumatization and creating gaps in documentation, while critical evidence is often collected late or inconsistently. The challenge is to design solutions that help capture and structure trauma-related accounts and evidence in ways that respect the realities of trauma while supporting reliable legal documentation.



Online education has grown rapidly, but many students in rural and low-income regions remain excluded due to unreliable or limited internet connectivity. Most live-class platforms depend on continuous video streaming, which fails on slow networks or expensive data plans, causing frequent disruptions and loss of access to lessons. As a result, students with 2G/3G or intermittent connections struggle to participate in real-time learning. The challenge is to create solutions that enable reliable participation in online education under low or unstable bandwidth while minimizing data usage and maintaining continuity during connectivity disruptions.

